

GOPIKANTH TIRUMANI

Senior NLP Engineer | Generative AI & Agentic AI Engineer

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SUMMARY

Senior NLP Engineer with 1.5+ years of experience designing and developing production-ready Generative AI, Agentic AI, RAG, and machine-learning systems. Hands-on expertise in LangGraph, LangChain, Model Context Protocol (MCP), multi-agent orchestration, hybrid retrieval, LLM tool calling, and FastAPI-based AI services. Experienced in designing end-to-end AI workflows, integrating business tools through controlled interfaces, and deploying AI solutions across AWS and GCP with observability and performance monitoring.

TECHNICAL SKILLS

- Generative AI & Agentic AI:** Multi-Agent Systems, LangGraph, LangChain, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), LLM Tool Calling, Prompt Engineering, Knowledge Graph RAG
- Languages & Backend:** Python, FastAPI, Pydantic, WebSockets, SQL, REST APIs
- Retrieval & Data:** FAISS, OpenAI Embeddings, BM25/TF-IDF, Redis, Vector Search
- Machine Learning:** XGBoost, LightGBM, CatBoost, Scikit-learn, PySpark, Time-Series Forecasting
- Cloud & Infrastructure:** AWS, GCP (BigQuery, Cloud Functions, Cloud Storage), Docker, LangSmith, Git

PROFESSIONAL EXPERIENCE

Senior NLP Engineer — RedFerns Tech

2025 – Present

- Design and develop production-ready Generative AI and Agentic AI applications across real-estate, education, and enterprise workflow domains.
- Architect multi-agent and RAG systems using LangGraph, LangChain, FAISS, OpenAI embeddings, BM25/TF-IDF, and Redis.
- Build MCP-based tool integrations with Pydantic-validated schemas for controlled interaction between LLM agents and business systems.
- Develop FastAPI and WebSocket-based AI services along with machine-learning pipelines for customer lifetime value, churn prediction, and time-series forecasting.
- Deploy and support cloud-based AI solutions across AWS and GCP, using Docker and LangSmith for deployment, tracing, debugging, and performance monitoring.

KEY PROJECTS

Multi-Agent Real Estate AI Platform

Tech Stack: Python, LangChain, LangGraph, FastAPI, MCP, FAISS, Redis, LangSmith, AWS, Docker

- Built a multi-agent real estate AI platform with specialized Broker Assistant, Sales Assistant, and On-Call Broker Assistant agents supporting CRM operations, lead qualification, property discovery, and sales workflows.
- Architected LangGraph-based agentic workflows with MCP integrations for CRM management, property search, scheduling, market analytics, and automated business operations.
- Implemented RAG and semantic matching using FAISS with Redis caching for objection handling, property recommendations, market intelligence, and conversational context.
- Implemented PII masking and secure tool execution across agent workflows, while developing real-time WebSocket-based on-call assistance with Docker and LangSmith for deployment, tracing, and observability.

Multilingual Educational RAG Platform

Tech Stack: Python, LangChain, LangGraph, FAISS, FastAPI, AWS, Redis, LangSmith

- Built a document-grounded RAG assistant with LangGraph-based intent routing for resource discovery, translation, and pronunciation assistance.
- Implemented automated FAISS embedding-refresh pipelines to keep the retrieval layer synchronized with updated knowledge sources.

- Added Redis caching for frequently requested RAG, translation, and text-to-speech responses, with LangSmith tracing for workflow monitoring.
- Built an automated student-analytics system generating LLM-based performance summaries, reducing reporting effort by ~95%, improving response time by ~40%, and increasing engagement by ~35%.

Agentic Development Environment

Tech Stack: Python, LangGraph, LangChain, MCP, FastMCP, FastAPI, WebSockets, Docker, Git, LangSmith

- Built a multi-agent agentic development environment using LangGraph with specialized agents for task planning, repository analysis, code generation, review, testing, debugging, security validation, and execution.
- Developed an MCP-based tool ecosystem enabling agents to create and modify files, execute Python and shell scripts, inspect repositories, run tests, perform Git operations, and manage Docker-based workloads.
- Implemented human-in-the-loop execution controls with tool-level permissions, command validation, approval gates, and sandboxed execution for sensitive system and repository operations.
- Designed closed-loop agent workflows that plan tasks, execute actions, analyze results, recover from failures, and iteratively validate outcomes, with WebSocket-based real-time execution updates and LangSmith tracing.

Customer Lifetime Value Prediction Pipeline

Tech Stack: Python, LightGBM, GCP, BigQuery, Cloud Functions, Cloud Storage, Pandas, NumPy, Joblib

- Built an event-driven GCP machine-learning pipeline to predict customer lifetime value across 30-day, 90-day, 180-day, and 13-month horizons.
- Developed a hybrid LightGBM classification + regression architecture, combining payment probability with predicted revenue to generate long-term customer value forecasts.
- Integrated BigQuery, Cloud Functions, and Cloud Storage for data processing, model artifacts, and scheduled pipeline execution, maintaining forecast calibration within approximately 90–110% across prediction horizons.

EDUCATION

Bachelor of Technology (B.Tech) — Lakireddy Bali Reddy College of Engineering **2021 – 2025**
CGPA: 8.99 / 10

CERTIFICATIONS

- Salesforce Certified AI Associate — Salesforce
- Generative AI Fundamentals — Databricks
- Introduction to Generative AI — Google Cloud
- Generative AI for Developers — Google Cloud

PUBLICATIONS

Smart Glasses with Voice Assistance and GPS for Independent Mobility of the Blind People
IEEE ICECCC 2025 · DOI: [10.1109/ICECCC65144.2025.11064275](https://doi.org/10.1109/ICECCC65144.2025.11064275) · [IEEE Xplore](#)